Aim: To perform statistical analysis and hypothesis testing using the Z-Test on a dataset in python

Software Required

- Python 3.x
- Jupyter Notebook
- Libraries: pandas, numpy, scikit-learn, matplotlib

Q1. What is Hypothesis Testing?

Hypothesis testing is a statistical method used to make decisions about a population based on sample data by testing an assumption (null hypothesis) against an alternative.

Hypothesis testing compares two opposite ideas about a group of people or things and uses data from a small part of that group (a sample) to decide which idea is more likely true. $z = \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}$

Q2. What is Z-Test?

A Z-test is a type of hypothesis test that compares the sample's average to the population's average and calculates the Z-score and tells us how much the sample average is different from the population average by looking at how much the data normally varies.

The first step in a Z-test is to state the null hypothesis, *H_0*.

This is what you believe to be true from the population, which could be the mean of the population, *μ_0*:

$$H_0 : \mu = \mu_0$$

Next, state the alternate hypothesis, *H_1*.

This is what you observe from your sample. If the sample mean is different from the population's mean, then we say the mean is not equal to μ_0:

$$H_1 : \mu \neq \mu_0$$

Q3. What is Train-Test Split?

Train-test split is a crucial machine learning technique that divides a dataset into two distinct subsets—typically 70-80% for training and 20-30% for testing—to evaluate a model's performance on unseen data.

Q4. Perform a Z-test on a game dataset to evaluate the following hypotheses:

- **H_0: The mean sales of Action-Adventure games equals the overall mean sales**
- **H_1: The mean sales of Action-Adventure games is greater than the overall mean sales**

Use a 75%–25% train-test split and interpret the results at α = 0.05.

Dataset:

id	name	developer	genre	release_year	metacritic_score	sales_millions	peak_players
1	Call of Duty 4: Modern Warfare	Infinity Ward	First-Person Shooter	2007	94	15.7	50000
2	Call of Duty: Modern Warfare 2	Infinity Ward	First-Person Shooter	2009	94	25	120000
3	Call of Duty: Black Ops	Treyarch	First-Person Shooter	2010	87	30.7	150000
4	Call of Duty: Warzone	Infinity Ward	Battle Royale	2020	80	0	6000000
5	Call of Duty: Modern Warfare (2019)	Infinity Ward	First-Person Shooter	2019	80	41	500000
6	Resident Evil 2	Capcom	Survival Horror	1998	93	4.96	0
7	Resident Evil 4	Capcom	Survival Horror	2005	96	7.1	0
8	Resident Evil 7: Biohazard	Capcom	Survival Horror	2017	86	12.4	20000
9	Resident Evil Village	Capcom	Survival Horror	2021	84	8.5	100000
10	Resident Evil (Remake)	Capcom	Survival Horror	2002	91	1.35	0
11	Grand Theft Auto V	Rockstar Games	Action-Adventure	2013	97	195	500000
12	Red Dead Redemption 2	Rockstar Games	Action-Adventure	2018	97	65	120000
13	Grand Theft Auto: San Andreas	Rockstar Games	Action-Adventure	2004	93	27.5	0
14	Bully	Rockstar Games	Action-Adventure	2006	87	1.5	0
15	Max Payne 3	Rockstar Games	Third-Person Shooter	2012	87	4	5000
16	God of War	Santa Monica Studio	Action-Adventure	2018	94	23	0
17	The Last of Us	Naughty Dog	Action-Adventure	2013	95	20	0
18	Uncharted 4: A Thief's End	Naughty Dog	Action-Adventure	2016	93	16	0
19	Spider-Man	Insomniac Games	Action-Adventure	2018	87	20	0
20	Horizon Zero Dawn	Guerrilla Games	Action RPG	2017	89	20	0
21	The Witcher 3: Wild Hunt	CD Projekt Red	RPG	2015	93	50	103000
22	Cyberpunk 2077	CD Projekt Red	RPG	2020	86	25	830000
23	Elden Ring	FromSoftware	Action RPG	2022	96	25	953000
24	Sekiro: Shadows Die Twice	FromSoftware	Action-Adventure	2019	90	8	124000
25	DOOM Eternal	id Software	First-Person Shooter	2020	88	12	75000
26	Portal 2	Valve	Puzzle	2011	95	12	40000
27	Left 4 Dead 2	Valve	Survival Horror	2009	89	20	50000
28	Baldur's Gate 3	Larian Studios	RPG	2023	96	15	875000
29	Divinity: Original Sin 2	Larian Studios	RPG	2017	93	7	50000
30	Hades	Supergiant Games	Roguelike	2020	93	4	47000
31	Hollow Knight	Team Cherry	Metroidvania	2017	90	6	18000
32	Stardew Valley	ConcernedApe	Simulation	2016	89	30	98000
33	Terraria	Re-Logic	Sandbox	2011	88	45	162000
34	Minecraft	Mojang	Sandbox	2011	93	238	230000
35	Among Us	InnerSloth	Party	2018	85	100	438000
36	Valheim	Iron Gate Studio	Survival	2021	91	12	500000

Code:

```
import pandas as pd
import numpy as np
from scipy.stats import norm
from sklearn.model_selection import train_test_split

data = pd.read_excel("EXP8.xlsx")

cols =
["id","name","developer","genre","release_year","metacritic_score","sales_millions","peak_
players"]
df = pd.DataFrame(data, columns=cols)

df = df[df["sales_millions"] > 0]

train_df, test_df = train_test_split(df, test_size=0.25, random_state=42)

print("H0: Mean sales of Action-Adventure games = Overall mean sales")
print("H1: Mean sales of Action-Adventure games > Overall mean sales\n")

population_mean = train_df["sales_millions"].mean()

sample = train_df[train_df["genre"] == "Action-Adventure"]["sales_millions"]
```

```
sample_mean = sample.mean()
sample_std = sample.std(ddof=1)
n = len(sample)

z_score = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
p_value = 1 - norm.cdf(z_score)

alpha = 0.05

print("===== TRAINING SET RESULTS =====")
print(f"Sample size (n): {n}")
print(f"Sample mean: {sample_mean:.2f}")
print(f"Population mean: {population_mean:.2f}")
print(f"Z-score: {z_score:.4f}")
print(f"P-value: {p_value:.6f}\n")

output_table = pd.DataFrame({
    "value1": [round(sample_mean, 6)],
    "value2": [round(population_mean, 6)],
    "score": [round(z_score, 6)],
    "p_value": [round(p_value, 6)],
    "Hypothesis_accepted": [
        "Yes" if p_value >= alpha else "No"
    ]
})

print(output_table.to_string(index=False))

if p_value < alpha:
    print("\nDecision: Reject H0")
else:
    print("\nDecision: Fail to Reject H0")

test_population_mean = test_df["sales_millions"].mean()
test_sample = test_df[test_df["genre"] == "Action-Adventure"]["sales_millions"]

if len(test_sample) > 0:
    test_mean = test_sample.mean()
    print("\n===== TEST SET CHECK =====")
    print(f"Test sample mean: {test_mean:.2f}")
    print(f"Test population mean: {test_population_mean:.2f}")

import matplotlib.pyplot as plt

train_size = len(train_df)
test_size = len(test_df)

labels = ["Training (75%)", "Testing (25%)"]
sizes = [train_size, test_size]

plt.figure()
plt.bar(labels, sizes)

plt.title("Train-Test Split (75% - 25%)")
plt.xlabel("Dataset Split")
plt.ylabel("Number of Samples")
```

```
for i, v in enumerate(sizes):
    plt.text(i, v, str(v), ha='center', va='bottom')

plt.show()
```

Output:

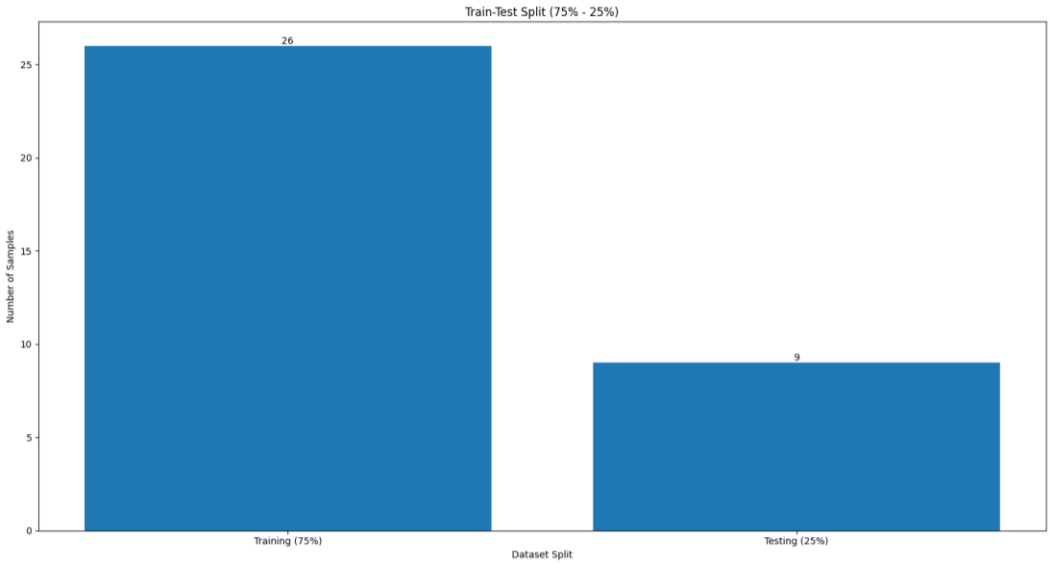
```
spectre@ghouled-gadget MINGW64 /d/degree/SEM 6/AIDS-1/Experiments/EXP8 (master)
$ py app.py
H0: Mean sales of Action-Adventure games = Overall mean sales
H1: Mean sales of Action-Adventure games > Overall mean sales

===== TRAINING SET RESULTS =====
Sample size (n): 7
Sample mean: 50.64
Population mean: 38.96
Z-score: 0.4670
P-value: 0.320249

   value1    value2    score  p_value Hypothesis_accepted
50.642857 38.956154 0.467003 0.320249                 Yes

Decision: Fail to Reject H0

===== TEST SET CHECK =====
Test sample mean: 10.75
Test population mean: 14.98
```



Conclusion:

Successfully applied the Z-test to evaluate hypotheses using dataset splitting, enabling statistically valid conclusions and demonstrating the role of hypothesis testing in data-driven decision-making.